MANCHESTER 1824
The University of Manchester

Introduction to Health Data Science

Group 3 Dashboard Presentation

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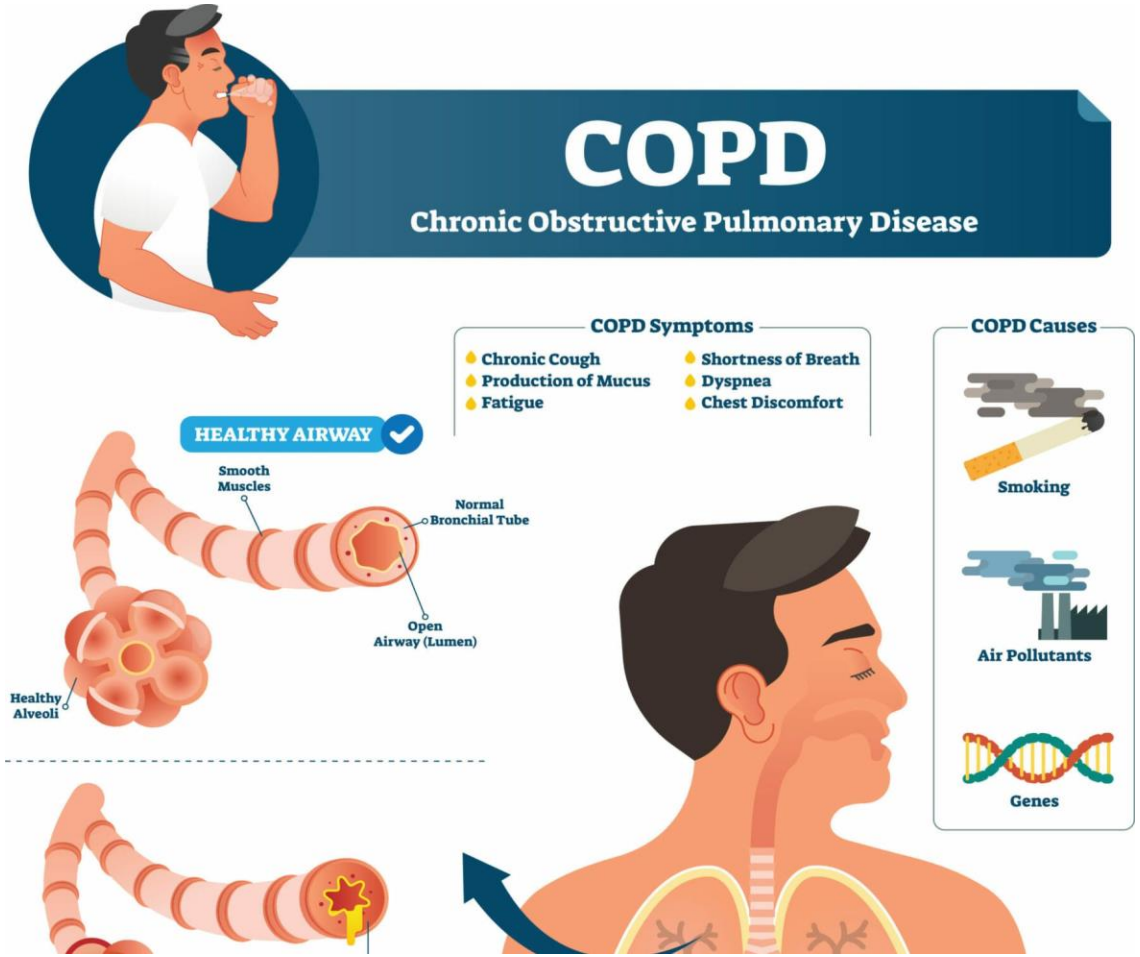
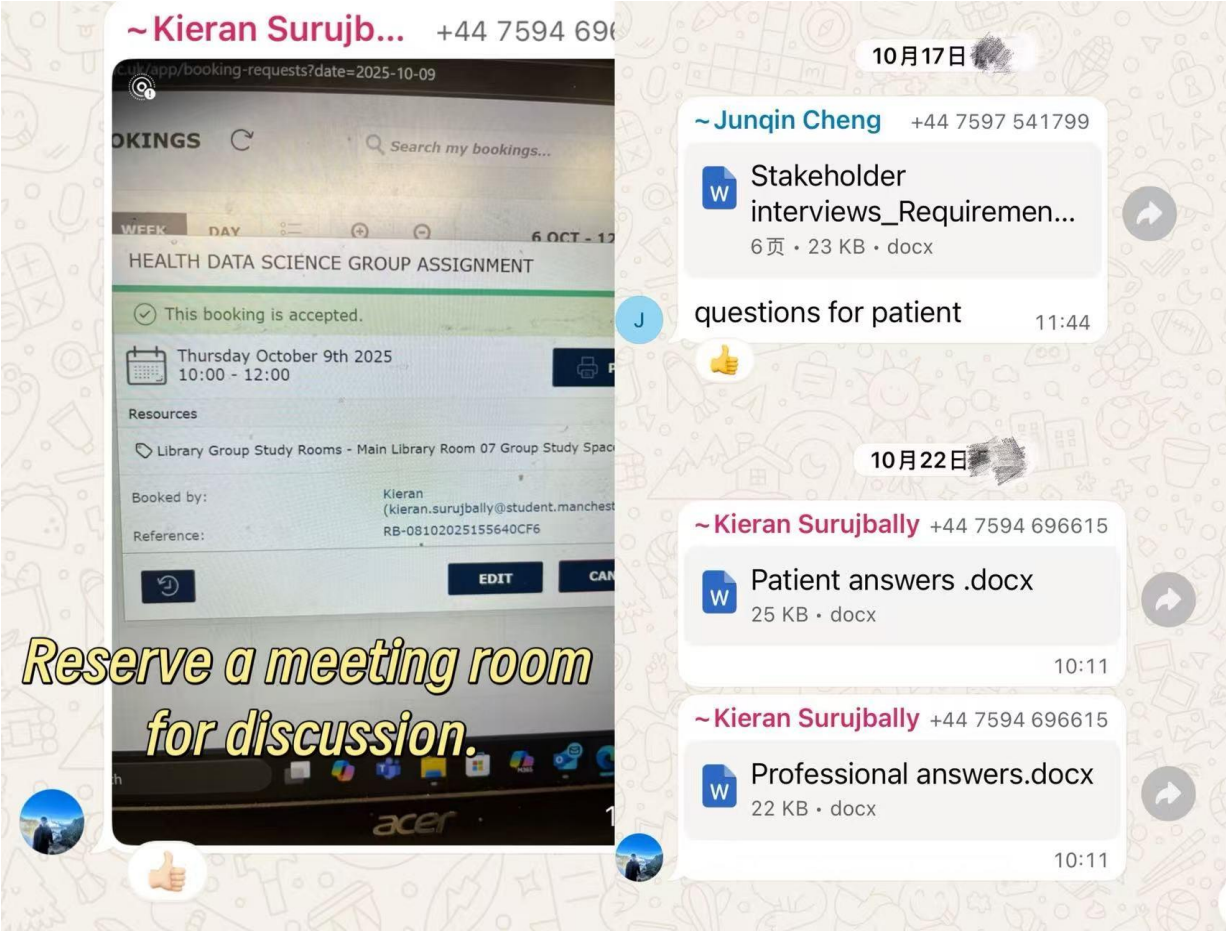
1.A brief introduction of COPD

- COPD is the name given to a group of lung conditions that cause breathing difficulties
- Mainly affects older adults who smoke
- Breathing problems tend to get gradually worse over time and can limit normal activities
- NO CURE! But treatments can help keep the condition under control.



2.Group Collaboration

- Discussion

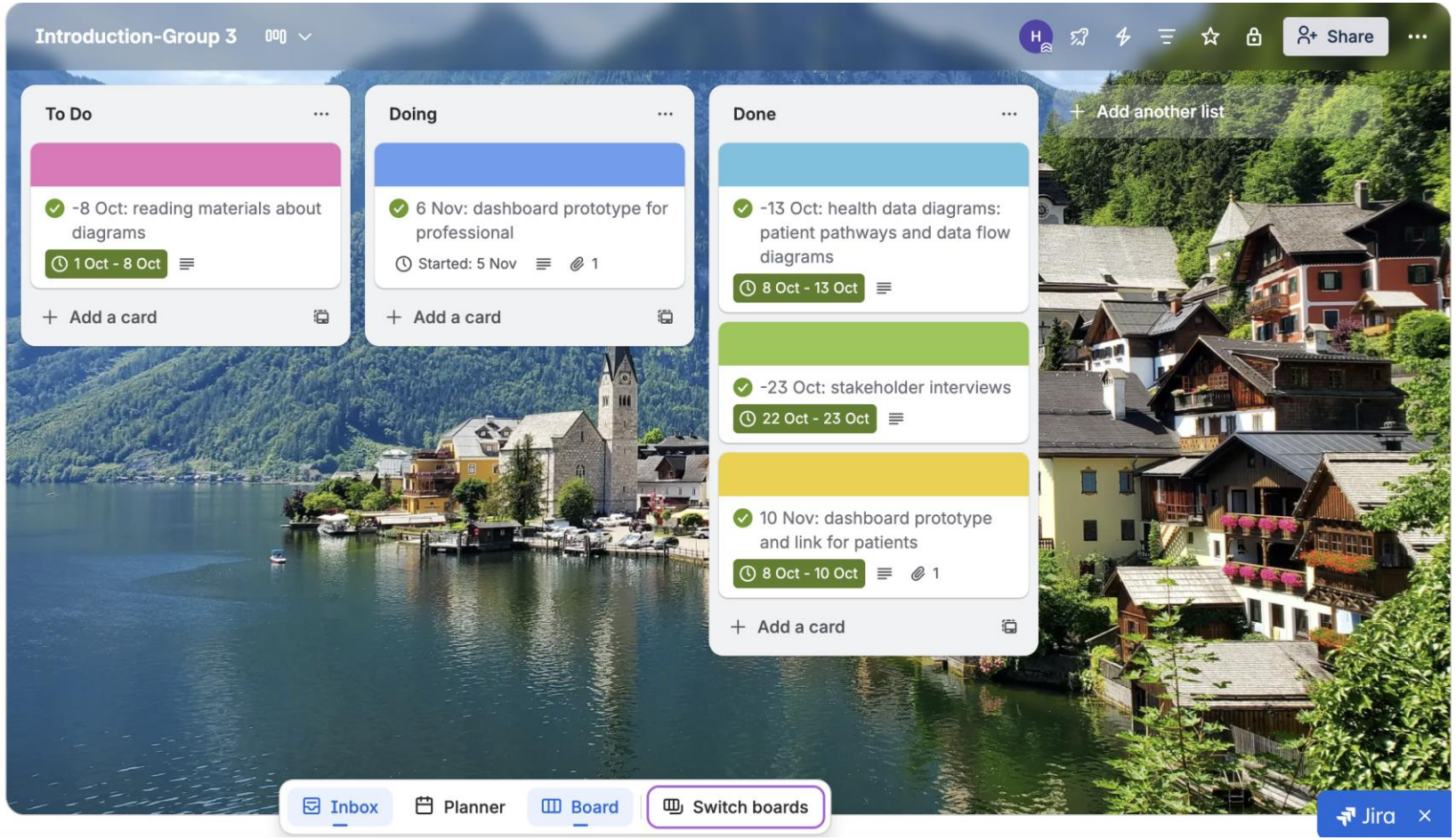


- Collaboration

CONTENT	PEOPLE
health data diagrams: patient pathway, data flow diagram and entity relationship diagram	Darvin Sureskumar, Kieran Surujbally, Bizhu Qian, Tianze Wang, Junqin Cheng
stakeholder interviews	Kieran Surujbally, Bizhu Qian, Tianze Wang, Junqin Cheng
dashboard prototype and development for patients	Darvin Sureskumar, Kieran Surujbally
dashboard prototype and development for professional	Bizhu Qian, Tianze Wang, Junqin Cheng

2.Group Collaboration

- Project Management



Data used

Provided datasets	External datasets
Full dataset : - Achievement score - NHS organisational structure (region, ICB, etc) - Admission and COPD prevalence - PCA rates	Oxygen and step count - Blood oxygen - Step count
	Air pollution and weather - Ozone level (O3) - Particulate matter PM2.5

3.Professional--Stakeholder Requirements

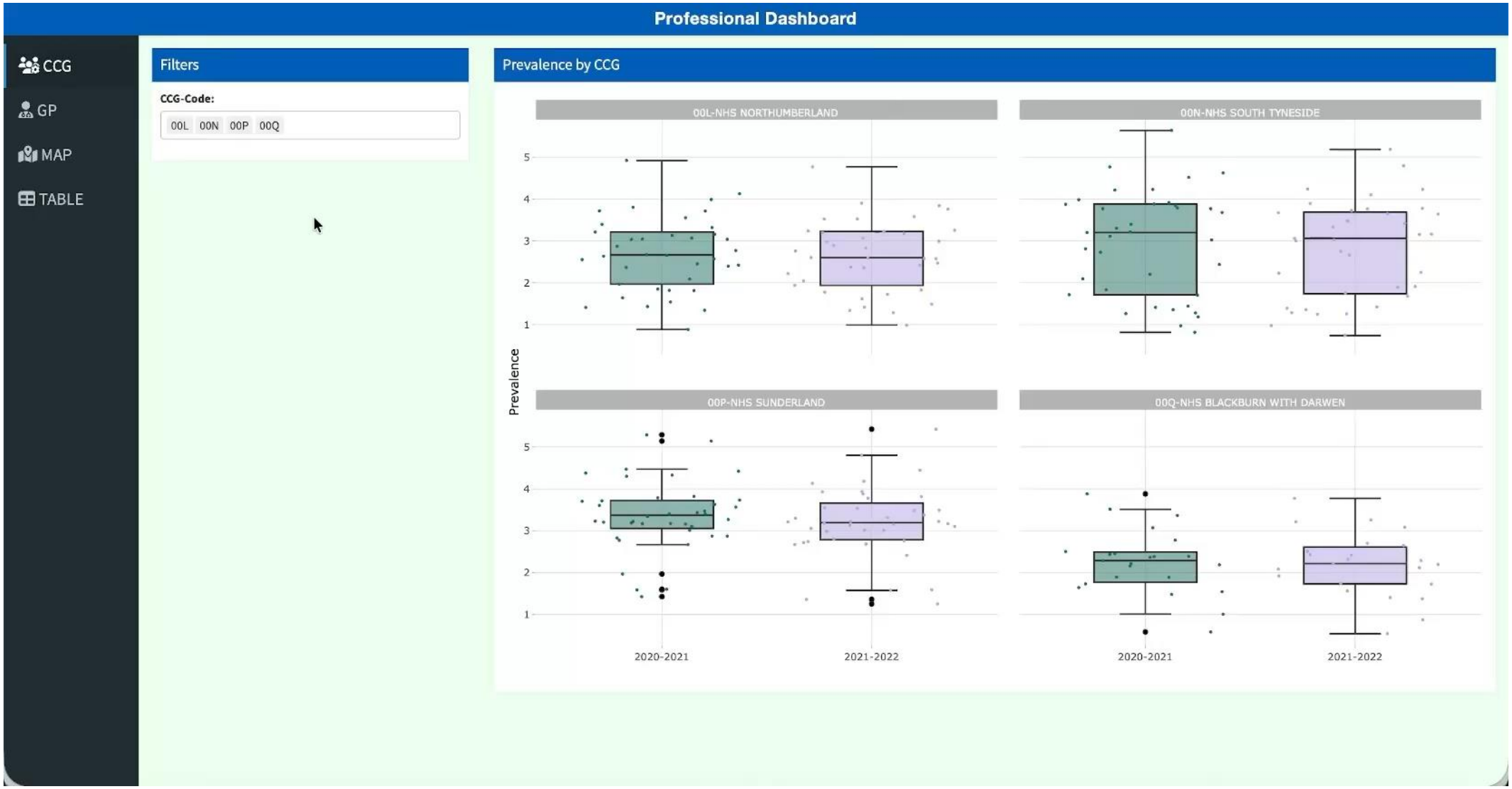
General information:

- Key metrics such as discharge rate and patient satisfaction and identify areas for improvement.
- Plans to include filters to extract specific data.
- Intends to use interactive tools, such as a map, to visualize specific metrics.
- Incidence of COPD, patients referral rate, good healthcare
- Filter

How should this be presented?

- User friendly, colorful, informatic
- The dashboard should be clear, efficient, and easy to interpret at a glance.
- Simply and easy

3.Professional--Dashboard Demonstration



4.Patient--Stakeholder Requirements

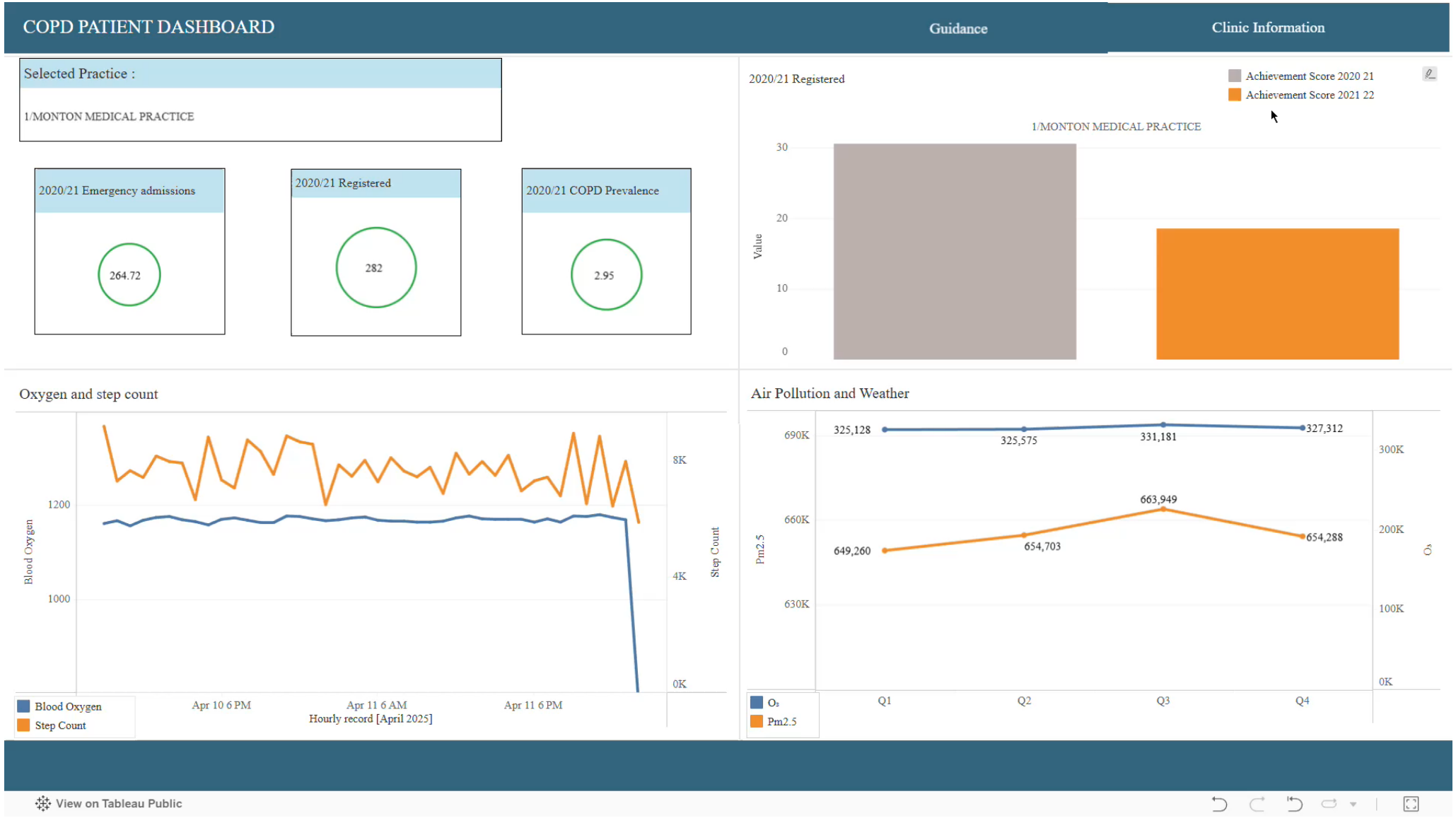
General information:

- Wants to know how often to get a flu vaccination and what to do during a flare-up
- Needs to know who to contact for help.
- Information on salbutamol treatment plan.
- Would like a specialist to review smartwatch data (walking, oxygen level).
- Weather/pollution levels.

How should this be presented?

- Prioritising simplicity
- Trend lines are preferred

4.Patient--Dashboard Demonstration



5.Reflection

- **Platform selection:**
- Tableau – easier navigation, lacks flexibility;
- RShiny – flexible, customizable, but complex.

- **Data Limitations:**
- Not able to complete requirement due to lack of data in the right format.

- **Improvements:**
- Optimize sizing, enhance loading speed via caching and modularization.

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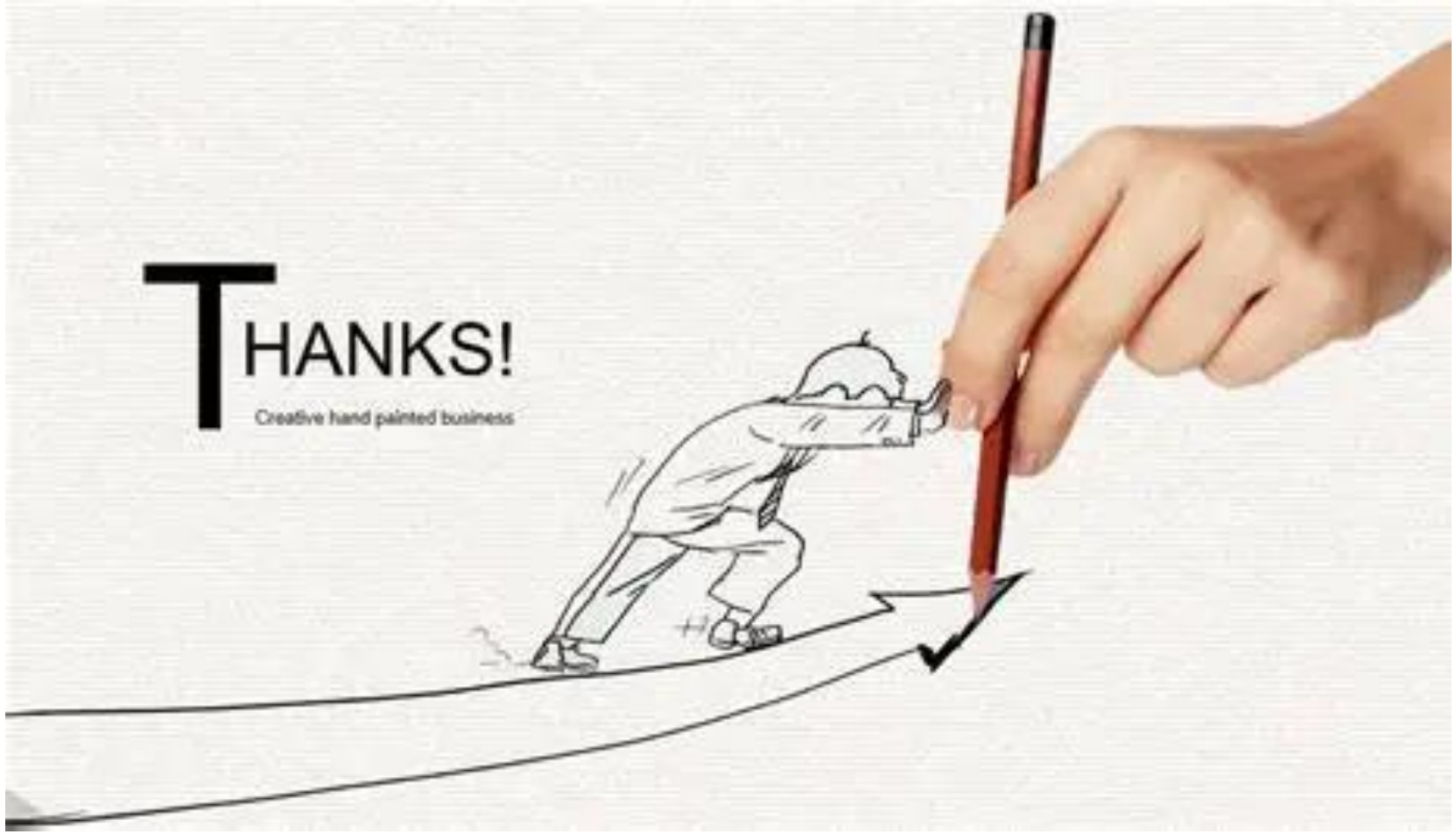
Thank you!